

HANA WAEL

Full-Stack Developer | Computer Science Student

Alexandria, Egypt • hanawaeltag12@gmail.com

LinkedIn: [linkedin.com/in/hana-wael-3783a0431](https://www.linkedin.com/in/hana-wael-3783a0431) • GitHub: github.com/hxnx444

Portfolio: hxnx444.github.io/Portfolio-website

PROFESSIONAL SUMMARY

Detail-oriented Computer Science student and full-stack developer with hands-on experience building web applications with React, Node.js and Flask, and a growing focus on AI/ML engineering. Focused on clean UI, accessible design, and writing maintainable, well-structured code. Proven ability to take end-to-end ownership of projects, from ideation and architecture to final deployment, while thriving in feedback-driven environments.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), HTML5, CSS3, Python, Java, PHP, SQL
- **Frameworks & Libraries:** React, Node.js, Flask, JavaFX, scikit-learn, NumPy
- **AI / Algorithms:** Neural networks (MLP), supervised classification, BFS, A* search, automata theory
- **UI/UX & Design:** Figma, Adobe XD, Responsive Design, CSS Animations, UI/UX Fundamentals
- **Developer Tools:** Git, GitHub, Visual Studio Code, Chrome DevTools
- **Core Concepts:** Component Architecture, RESTful APIs, Object-Oriented Programming (OOP), Relational Databases

PROJECTS

Maze Agent | AI Maze Solver *Python, Flask, scikit-learn, JavaScript*

- Built a Flask web app where a scikit-learn MLP classifier predicts maze difficulty (easy / medium / hard) from five extracted features, reaching about 90% accuracy on a held-out test set of synthetic mazes.
- Implemented BFS and A* search plus an agent that picks between them from the prediction and explains its choice, with live visualization in the browser.
- Documented the limits (synthetic labels from a scoring formula) and planned training on real node counts.

NFA → DFA Conversion Engine | Automata Theory *Java, Graphviz*

- Built a Java tool that converts regular expressions into minimized DFAs using Thompson's construction, subset construction and Hopcroft's minimization.
- Generated Graphviz DOT output to render state-machine diagrams automatically.

Aura Vision | Full-Stack Productivity Web App *React, Node.js, CSS Animations*

- Built a full-stack productivity app by connecting interactive React components to a Node.js backend, implementing dynamic state management and a scalable component architecture.
- Applied CSS animations and responsive design principles to deliver a polished, accessible user experience.

Library Management System | Desktop Application *Java, JavaFX, SQLite/MySQL*

- Engineered a desktop application with a robust OOP architecture to manage member registration, book tracking, and borrowing workflows for 500+ records.
- Resolved critical bugs in the return-book and borrow-list features and streamlined the user flow by removing redundant logout elements.
- Integrated a relational database for persistent data handling.

EDUCATION

Bachelor of Science in Computer Science (Expected Graduation: 2028), Alexandria University, Faculty of Science

- Relevant Coursework: Human-Computer Interaction (HCI), Data Structures & Algorithms, Advanced Programming, Computer Graphics, Web Development, System Programming, Theory of Computation, Discrete Structures, Object-Oriented Programming.

CERTIFICATIONS

- **ITIDA Freelance Certificate**, Information Technology Industry Development Agency (ITIDA), Egypt